

Pair Programming

Team-13

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Part 3: Object Detection (50 Points)

1. Explain what is Mask R-CNN (5 points)

Mask R-CNN is used for Enhancing Object Detection

Mask R-CNN extends Faster R-CNN by adding a third branch for object segmentation, allowing precise pixel-level outlining of objects.

Key Features:

- Simplicity: Easy to train and implement.
- Performance: Outperforms existing methods on various tasks.
- Efficiency: Adds minimal overhead to Faster R-CNN.
- Flexibility: Can be adapted to different tasks with ease.

Architecture:

- Builds upon Faster R-CNN, adding a mask prediction branch.
- Pixel-to-Pixel Alignment: Ensures accurate object segmentation.
- Framework: Follows a two-stage procedure, incorporating Region Proposal Network (RPN) and mask prediction.

Applications:

- OpenStreetMap Enhancement: Identifies features like sports fields in satellite images.
- Humanitarian Mapping: Helps create maps for relief efforts using satellite imagery.

2. Explain U-Net (5 points)

UNet, developed in 2015 at the University of Freiburg, Germany, is a powerful architecture for biomedical image segmentation.

Architecture:

- U-Shaped Design: UNet features a U-shaped encoder-decoder network.
- Encoder: It reduces spatial dimensions and increases feature channels through convolutional blocks.
- Skip Connections: Links between encoder and decoder blocks preserve crucial information.
- Bridge: Connects encoder and decoder, facilitating smooth information flow.
- Decoder: Upsamples feature maps and combines them with skip connections for detailed segmentation.
- Final Output: Sigmoid activation produces pixel-wise classification masks.

Working Principle:

1. Feature Extraction: The encoder extracts abstract features from input images.
2. Information Preservation: Skip connections maintain important details during downsampling.
3. Information Fusion: The bridge combines encoder features with decoder inputs.
4. Semantic Segmentation: The decoder refines features and generates detailed segmentation masks.
5. Output: Sigmoid activation produces masks representing object boundaries.

Conclusion: UNet's straightforward architecture and effective information flow make it a go-to solution for various biomedical image segmentation tasks.

3. What is intersection over union (IOU) (2 points)

Intersection over Union (IOU), also known as the Jaccard Index in computer science, is a metric widely used in object detection, segmentation, and image classification to evaluate the accuracy of a method. Calculating the ratio of the intersection area to the union area of two bounding boxes or segmentation masks is the method of determining if there is an overlap between two bounding boxes or segmentation masks.

Formula:

$$[IOU = \frac{\text{Area_of_Intersection}}{\text{Area_of_Union}}]$$

Interpretation:

- IOU ranges from 0 to 1.
- A value of 1 indicates perfect overlap between the two bounding boxes.
- A value of 0 indicates no overlap between the two bounding boxes.

Usage:

- In object detection, IOU is used to measure the accuracy of predicted bounding boxes compared to ground truth bounding boxes.
- In image segmentation, IOU is used to evaluate segmentation mask quality by comparing predicted masks with ground truth masks.

4. PreTrained Model

1. Downloaded the dataset from kaggle. Corrupted xml files and respective images were removed and loaded to notebook
2. Images were transformed and augmented in the notebook.
3. Prepared the train, valid, test split in notebook
4. Random images from train,test,plotted in notebook

We were unable to train the model using the above dataset, so we have chosen to annotate the images in roboflow and used it for training our custom model.

5. We have chosen a combination of Faster R-CNN with MobileNetV3 model to build our own model.

Faster R-CNN is an object detection algorithm that combines a region proposal network (RPN) and a classification network to efficiently detect objects in images.

When combined with the MobileNetV3 architecture, it creates an efficient and lightweight model for object detection tasks, especially on resource-constrained devices or in scenarios where computational power is limited.

Architecture:

Faster R-CNN:

- Faster R-CNN is an extension of the R-CNN (Region-based Convolutional Neural Network) family of object detectors.

- It introduces an RPN that efficiently proposes regions of interest (RoIs) that are likely to contain objects.
- These RoIs are then processed by the classification network, which classifies the objects and refines the bounding box coordinates.
- This two-stage approach (region proposal and classification) makes Faster R-CNN efficient and accurate.

MobileNetV3:

- MobileNetV3 is a lightweight convolutional neural network architecture designed for mobile and embedded devices.
 - It is based on depth-wise separable convolutions, which reduce the computational cost and model size compared to traditional convolutions.
 - MobileNetV3 introduces additional optimization techniques, such as squeeze-and-excite blocks and hard swish activation functions, to further improve performance and efficiency.
6. Trained the model on the custom dataset for 300 epochs, learning rate = 0.01 and SGD optimizer. Weights were saved and uploaded and validated.
 7. Got Accuracy of 80% which is considered as good accuracy for object detection. This accuracy score means that the model correctly detects and classifies objects in 80% of the instances. It demonstrates our model's ability to precisely localize objects within the images and assign the correct class labels to them. The fact that this accuracy is achieved with the combination of Faster R-CNN and MobileNetV3 architecture indicates that the model is capable of delivering good performance while being computationally efficient and lightweight.
 8. Predictions were run on test data and predicted images and bbox were saved.
 9. IoU was calculated and it was 0.7 which indicates that the predicted bounding boxes have a substantial overlap with the ground truth bounding boxes. On average, the predicted bounding boxes overlap with the ground truth bounding boxes by 70%. Even though its a good score the model can still be improved to increase the performance.
 10. Printed few predicted images in which the model was able to detect the objects and bounding boxes accurately.

In FasterRCNN folder, codefiles, model weights and csv file is uploaded.

5. YOLOv8 model

1. Downloaded the dataset from kaggle. Corrupted files were removed and transformed and augmented using roboflow and the final images set is around ~2000 images.
2. Prepared the train, valid, test split in notebook
3. Random images from train,test,plotted in notebook
4. Downloaded YOLOv8 model and data.yaml is prepared.
5. Trained the YOLOv8 model on the custom dataset and weights and metrics were saved
6. Best weight model is loaded and performed the validation
7. Got meanAveragePrecision50 as 0.707 which signifies a strong performance in accurately detecting and localizing objects across various classes. On average the model correctly detects 70% of the ground truth objects with a predicted bounding box that has at least 50% overlap (IoU ≥ 0.5) with the ground truth bounding box. This indicates that the model is quite effective in

identifying and precisely localizing objects within the specified IoU threshold of 0.5. An mAP50 of 0.7 is generally considered a good result, as it demonstrates the model's ability to reliably detect and localize a substantial portion of objects with reasonable bounding box accuracy.

8. Predictions were run on test data and images and predicted boxes were saved in predicts folder.
9. IoU was calculated and it was 0.699 which signifies that, on average, the predicted bounding boxes have a significant overlap with the ground truth bounding boxes. Specifically, it means that the predicted bounding boxes overlap with the ground truth boxes by approximately 69.9%. An IoU score above 0.5 is generally considered acceptable, and a score close to 0.7 indicates that the model is effectively localizing objects with a high degree of accuracy in terms of bounding box placement and dimensions.
10. Graphs, correlation matrix were plotted from the results
11. Printed few predicted images in which the model was able to detect the objects and bounding boxes accurately.

In YOLO folder, dataset contains all the 3 folders data used for training and validation. Weight files, evaluation metrics, csv files, predicted images, and Bbox coordinate files are stored in 'model' folder in YOLO.